

5054/22/M/J/25 - May/June 2025

Theory: Mark Scheme & Model Answers

5054/22/M/J/25 - Q1

Mark Scheme

Question	Answer	Marks
1(a)(i)	points plotted to within 1/2 a small square	B1
	smooth curve drawn	B1
1(a)(ii)	(average speed =) d/t in any form	C1
	195 (cm/s)	A1
1(a)(iii)	draw a tangent (at $t = 0.40$ s)	B1
	find the gradient / slope (of the graph)	B1
	or the gradient of the tangent	
1(b)(i)	mention of air resistance / resistive force	B1
	resultant force decreases	B1
	or air resistance / upwards force increases (as speed increases) / becomes high(er)	
	(eventually) air resistance = weight (and acceleration is zero) or air resistance balances / cancels weight	B1
1(b)(ii)	straight line (of positive) or (line of) constant gradient / slope	B1

Exemplar Key

(a)(i) Plotting the Graph

Plot each coordinate pair from Table 1.1 carefully on the grid, ensuring points are within half a small square of their exact positions. Then, draw a single smooth curve passing through these plotted points.

(a)(ii) Average Speed

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

At $t = 0.40$ s, the distance fallen is 78 cm.

$$v = \frac{78}{0.40}$$

$$\text{speed} = \boxed{195 \text{ cm/s}}$$

(a)(iii) Instantaneous Speed

To find the speed at exactly $t = 0.40$ s, draw a straight tangent line touching the curve at the point where $t = 0.40$ s. Then, calculate the gradient (slope) of this tangent line.

(b)(i) Changing Acceleration

As the ball falls and its speed increases, the upward force of air resistance acting against it also increases. This increasing upward force opposes the constant downward weight of the ball, meaning the overall downward resultant force decreases (causing acceleration to decrease). Eventually, the upward air resistance fully balances the downward weight, making the resultant force zero, which means the acceleration becomes zero.

(b)(ii) Graph with Zero Acceleration

When acceleration is zero, the ball moves at a constant terminal velocity, so the distance-time graph becomes a straight line with a constant positive gradient.

5054/22/M/J/25 - Q2

Mark Scheme

Question	Answer	Marks
2(a)	work done or energy per unit time	B1
2(b)	(input power =) power output / efficiency in any form 860 (W)	C1 A1
2(c)(i)	(work =) force $\times$ distance or mgh 1100 (J)	C1 A1
2(c)(ii)	(time =) work / (output) power 1.9 or 1.8 s	C1 A1
2(d)	Any one of: friction/air resistance electrical heating	B1

Exemplar Key

(a) Definition of Power

Power is the work done or energy transferred per unit time.

(b) Input Power

$$\begin{aligned}\text{efficiency} &= \frac{\text{power output}}{\text{power input}} \\ 0.70 &= \frac{600}{\text{power input}} \\ \text{power input} &= \frac{600}{0.70} = 857.14 \dots \\ \text{input power} &= \boxed{860 \text{ W}}\end{aligned}$$

(c)(i) Work Done

$$\begin{aligned}\text{work} &= \text{force} \times \text{distance} = mg \times h \\ \text{work} &= 50 \times 9.8 \times 2.3 = 1127 \\ \text{work done} &= \boxed{1100 \text{ J}}\end{aligned}$$

(c)(ii) Time Taken

$$P = \frac{\text{work}}{\text{time}}$$

$$t = \frac{1127}{600} = 1.878 \dots$$

$$\text{time taken} = \boxed{1.9 \text{ s}}$$

(d) Thermal Energy Transfer

Energy is transferred to thermal energy due to friction between the moving mechanical parts of the fork-lift truck.

5054/22/M/J/25 - Q3

Mark Scheme

Question	Answer	Marks
3(a)(i)	($P =$) force / area in any form	C1
	1.9×10^5 (Pa)	A1
3(a)(ii)	(increasing pressure) decreases the area	B1
3(b)(i)	mass (of the particle) $\times$ velocity (of the particle)	B1
3(b)(ii)	on collision: change of momentum/direction / velocity or on collision: momentum transferred to wall	B1
	force related to change in momentum / time or force of equal size on tyre / wall and on particle (Newton's 3rd law)	B1
3(c)	(particles have) high(er) speed/ velocity/momentum/kinetic energy	B1
	more collisions per second	B1
	or hit (wall) hard(er) or larger change in momentum per sec	

Exemplar Key

(a)(i) Pressure Exerted

$$P = \frac{F}{A}$$

$$P = \frac{1350}{0.0070} = 192857.1 \dots$$

$$\text{pressure} = \boxed{1.9 \times 10^5 \text{ Pa}}$$

(a)(ii) Change in Area

Since the force (weight) remains constant, increasing the internal pressure means the area of the tyre in contact with the road decreases.

(b)(i) Definition of Momentum

Momentum is the product of the mass of a particle and its velocity.

(b)(ii) Force on the Wall

When an air particle collides with the tyre wall and bounces off, it undergoes a change in velocity, and thus a change in momentum. The rate of change of momentum exerts a force on the particle, and according to Newton's third law, an equal and opposite force is exerted by the particle onto the wall of the tyre.

(c) Pressure Increase with Temperature

As temperature rises, the air particles gain kinetic energy and move at higher speeds. Because they are moving faster, they hit the tyre walls harder (experiencing a larger change in momentum per collision) and collide with the walls more frequently. This increased rate of change of momentum over the tyre's area results in a higher pressure.

5054/22/M/J/25 - Q4

Mark Scheme

Question	Answer	Marks
4(a)	motion of particles: backwards and forwards / to and fro / closer and further apart	B1
	(oscillation) in the direction of travel of the wave / to the right/in direction of microphone/longitudinal / horizontal / parallel to vibrations of tuning fork	B1
	or mention of compressions and rarefactions	
4(b)(i)	number of oscillations per second or number of wave (length)s (passing a point) per second	B1
4(b)(ii)	2.5 (waves) seen or (1 wave takes) 0.02 (s) or ($f =$) $1/T$ or (number of) waves / time	C1
	50 (Hz)	A1
4(b)(iii)	quieter / less volume	B1
4(b)(iv)	half frequency shown and at least one complete oscillation seen	B1
4(c)(i)	(at least) three correctly curved wavefronts centered on gap	B1
	same wavelength as on left and some curving in correct direction	B1
4(c)(ii)	diffraction	B1

Exemplar Key

(a) Motion of Air Particles

The air particles oscillate backwards and forwards (to and fro) parallel to the direction of travel of the wave. This longitudinal motion creates alternating regions of compressions and rarefactions.

(b)(i) Meaning of Frequency

Frequency is the number of complete oscillations or wavelengths passing a point per second.

(b)(ii) Calculating Frequency

From the graph, exactly 2.5 complete waves are shown over a time interval of 0.050 s.

$$f = \frac{\text{number of waves}}{\text{time}} = \frac{2.5}{0.050}$$

$$\text{frequency} = \boxed{50 \text{ Hz}}$$

(b)(iii) Change in Sound

Because the amplitude decreases, the sound becomes quieter (lower volume).

(b)(iv) New Oscilloscope Trace

Draw a sine wave starting from the same zero point with the exact same peak amplitude, but stretched horizontally so that it completes only 1.25 waves across the entire grid (since half the frequency means the waves are twice as wide).

(c)(i) Drawing Wavefronts

On the right side of the gap, draw three semi-circular arcs centered at the gap. The radial distance between these curved wavefronts must remain equal to the original wavelength (the spacing between the straight vertical lines on the left).

(c)(ii) Name of Process

Diffraction

5054/22/M/J/25 - Q5

Mark Scheme

Question	Answer	Marks
5(a)	swap voltmeter for ammeter	B1
	have cells all facing the same way, e.g. reverse left two cells	B1
5(b)(i)	$(R =) V/I$ in any form, e.g. $6/0.8$	C1
	7.5 (Ω)	A1
5(b)(ii)	shorter and one of: resistance smaller or resistance proportional to length larger current (for same voltage) smaller voltage to produce (an equal) current	B1
5(b)(iii)	6 (.0 cm)	B1

Exemplar Key

(a) Changes to Circuit

1. Swap the positions of the voltmeter and the ammeter so that the ammeter is in series with the wire and the voltmeter is placed in parallel across it. 2. Reverse the orientation of the two cells on the left so that all four cells are facing the same direction (positive terminals all pointing the same way) to provide a non-zero electromotive force.

(b)(i) Resistance of 9.0 cm Wire

From line P on the graph, at a voltage of 6.0 V, the current is 0.80 A.

$$R = \frac{V}{I} = \frac{6.0}{0.80}$$

$$\text{resistance} = \boxed{7.5 \, \Omega}$$

(b)(ii) Length Comparison

Line Q represents a shorter wire. At any given voltage, line Q shows a larger current flowing than line P, meaning it has a smaller resistance. Because the resistance of a wire is directly proportional to its length, a smaller resistance must correspond to a shorter length.

(b)(iii) Length of Wire Q

From line Q, at a voltage of 4.0 V, the current is 0.80 A.

$$R_Q = \frac{4.0}{0.80} = 5.0 \, \Omega$$

Since length is proportional to resistance:

$$\frac{L_Q}{L_P} = \frac{R_Q}{R_P} \implies L_Q = 9.0 \times \frac{5.0}{7.5}$$

$$\text{length} = \boxed{6.0 \, \text{cm}}$$

5054/22/M/J/25 - Q6

Mark Scheme

Question	Answer	Marks
6(a)	$(I) = P/V$ or 2000/230 or 8.7 (A)	M1
	fuse rating 13 A underlined or circled	A1
	The fuse rating needs to be the lowest value above / slightly larger than the normal current/8.7 A	B1
	or 5/8 A fuse blows under normal operation / 8.7 A or 30 A fuse may not blow / heater or circuit damaged if fault or current larger than normal 8.7 A	
6(b)	fuse melts/blows/ or cuts off current/circuit	B1
6(c)(i)	0.95 (A)	B1
6(c)(ii)	17	B1
	Any two from: (trip switches can / are) be reset/used more than once faster acting/turns off immediately more sensitive be tested without replacement have an external indication that they are working can trip on difference in live and neutral currents	B2

Exemplar Key

(a) Choosing a Fuse

13 A Calculation:

$$I = \frac{P}{V} = \frac{2000}{230} = 8.695 \dots \text{ A}$$

Explanation: The normal operating current is approximately 8.7 A. The fuse rating must be slightly larger than this working current. A 5 A or 8 A fuse would blow under normal operation, cutting power to the heater. A 30 A fuse is too large and might not blow during a fault, risking damage or a fire.

(b) Function of a Fuse

When the current becomes too large, the thin wire inside the fuse becomes hot and melts (blows), which breaks the circuit and cuts off the current.

(c)(i) Current in Lighting Circuit Fuse

The total current is the sum of the currents in parallel branches:

$$I_{\text{total}} = 0.26 + 0.26 + 0.43 = 0.95$$

$$\text{current} = \boxed{0.95 \text{ A}}$$

(c)(ii) Maximum Additional Lamps

The lighting fuse is rated at 5 A. The two existing type P lamps take $0.26 + 0.26 = 0.52$ A. Current available for new lamps = $5.0 - 0.52 = 4.48$ A. Number of additional type P lamps = $4.48/0.26 = 17.23$.

$$\text{number of additional lamps} = \boxed{17}$$

(d) Advantages of Trip Switches

1. A trip switch can simply be reset and used multiple times, whereas a blown fuse must be fully replaced. 2. A trip switch is much faster acting, cutting off the current almost immediately when a fault is detected.

5054/22/M/J/25 - Q7

Mark Scheme

Question	Answer	Marks
7(a)	split ring commutator:	B1
	to reverse the current (in the coil)	
	reverse current/connections every half turn / twice in one turn / every 180° / when coil is vertical	B1
	brushes: to connect coil/commutator or (split) rings to battery / (external) circuit or to provide current to coil, commutator or (split) rings or to avoid wires (to coil) tangling / twisting	B1
7(b)	direction: (from left) to right/from magnet 1 to magnet 2	B1
	explanation:	B1
	current is from plus to minus / towards front/out of plane/page/outwards / top to bottom of wire / downwards along wire / towards the split ring on the left	
	left-hand rule mentioned and explained using two of current, field or motion / force	B1
7(c)	no change	B1
7(d)(i)	(moment =) force $\times$ distance in any form	C1
	0.69	A1
	Nm	B1
7(d)(ii)	(perpendicular) distance between (line of action of) forces decreases or (sides of) coil enter weaker magnetic field or coil leaves magnetic field	B1

Exemplar Key

(a) Purpose of Commutator and Brushes

Split-ring commutator: Reverses the direction of the current in the coil every half turn (when the coil is vertical) to ensure the coil continues rotating in the same direction. Brushes: Provide a sliding electrical contact that transfers current from the stationary battery to the rotating commutator without the connecting wires tangling.

(b) Magnetic Field Direction

direction: From left to right (from magnet 1 to magnet 2). explanation: Using Fleming's left-hand rule. On the left side of the coil, the conventional current flows outwards (towards the split ring), and the resulting force is upwards. For the thumb (force) to point up and the second finger (current) to point outwards, the first finger must point from left to right, identifying the magnetic field direction.

(c) Reversing Both Field and Current

There is no change to the rotation of the coil (it continues rotating in the same direction).

(d)(i) Total Moment

$$\text{total moment} = (\text{force} \times \text{distance})_{\text{left}} + (\text{force} \times \text{distance})_{\text{right}}$$

$$M = (2.3 \times 0.15) + (2.3 \times 0.15) = 0.345 + 0.345 = 0.69$$

$$\text{moment} = \boxed{0.69} \quad \text{unit} = \boxed{\text{Nm}}$$

(d)(ii) Change in Turning Effect

As the coil rotates from the horizontal to the vertical position, the perpendicular distance from the axis of rotation to the vertical lines of action of the forces decreases, resulting in a smaller turning effect (moment).

5054/22/M/J/25 - Q8

Mark Scheme

Question	Answer	Marks
8(a)	shielding: protects workers / absorbs radiation	B1
	control rods: controls number of neutrons / absorbs neutrons	B1
	moderator: slows down the neutrons	B1
8(b)	neutron hits / collides U-235 (nucleus)	B1
	nucleus splits/forms daughter nuclei	B1
	emits more neutrons	B1
8(c)(i)	amount of energy / heat needed for a unit rise in temperature	M1
	for unit mass	A1
8(c)(ii)	large (specific heat capacity) and one of: smaller rise in temperature / temp. stays lower heats up slower / longer time less chance of boiling smaller mass needed (for same temperature rise) / less flow rate / smaller pumps more energy absorbed for same temp. rise or small(er heat capacity) and larger temperature rise (to heat boiler more efficiently) / heats up faster	B1 B1

Exemplar Key

(a) Purpose of Reactor Parts

concrete shielding: Absorbs hazardous radiation, protecting the workers outside. control rods: Absorbs neutrons to control the rate of the fission chain reaction. moderator: Slows down fast-moving neutrons so they can be absorbed by uranium-235 nuclei.

(b) Fission of Uranium-235

A slow-moving neutron hits and is absorbed by a uranium-235 nucleus, causing the nucleus to become highly unstable. This unstable nucleus splits into two smaller daughter nuclei, releasing a large amount of energy and emitting two or three more fast-moving neutrons.

(c)(i) Definition of Specific Heat Capacity

Specific heat capacity is the amount of thermal energy required to raise the temperature of a unit mass (e.g., 1 kg) of a substance by 1°C (or 1 K).

(c)(ii) Choice of Coolant Capacity

The coolant should have a large specific heat capacity. This allows it to absorb a vast amount of thermal energy from the fuel rods while only experiencing a relatively small rise in temperature, preventing the coolant from boiling within the reactor core.

5054/22/M/J/25 - Q9

Mark Scheme

Question	Answer	Marks
9(a)	Mercury orbit inside Earth's orbit	B1
	Mars orbit between orbit of Earth and Jupiter	B1
9(b)	the Sun is in the way	B1
9(c)(i)	gravitational (force / attraction from Sun)	B1
9(c)(ii)	all four forces towards Sun by eye	B1
9(c)(iii)	increases (from 1) to 3 and decreases (from 3 to 4)	B1
	explanation of statement in description	B1
	e.g. (stronger force) because comet is close(r) to Sun (at position 3)	
9(d)	speed = $2\pi r/T$ in any form	C1
	time for Jupiter to orbit Sun = 3.8×10^8 (s)	C1
	or time for Earth to orbit Sun = 3.1×10^7 (s)	
	or distance increases by 7.8/1.5 and speed decreases by 30/13	
	or $\frac{7.8(\times 10^8)}{1.5(\times 10^8)} \times \frac{30}{13}$	
	number of orbits = 12	A1

Exemplar Key

(a) Drawing Orbits

Draw an approximately circular or slightly elliptical path completely inside the orbit of Earth, and label it "Mercury". Draw another orbit entirely between the orbit of Earth and the orbit of Jupiter, and label it "Mars".

(b) Visibility of Comet

When the comet is at position 3 and the Earth is at position 5, they are on exactly opposite sides of the Sun. Because the Sun is directly in the line of sight, the comet cannot be seen.

(c)(i) Name of Force

Gravitational force

(c)(ii) Direction of Force

From each of the numbered points (1, 2, 3, 4), draw a straight arrow pointing directly inward toward the central black dot representing the Sun.

(c)(iii) Strength of Force

description: The magnitude of the force increases as the comet moves from position 1 to position 3, and then decreases as it moves from position 3 to position 4. explanation: Gravitational attraction is stronger when objects are closer together. Position 3 is the closest point to the Sun, so the force is strongest there.

(d) Number of Orbits

The time for one orbit is $T = \frac{2\pi r}{v}$. For Earth:

$$T_E = \frac{2\pi \times 1.5 \times 10^8 \text{ km}}{30 \text{ km/s}} = \frac{3.0\pi \times 10^8}{30} = \pi \times 10^7 \text{ s}$$

For Jupiter:

$$T_J = \frac{2\pi \times 7.8 \times 10^8 \text{ km}}{13 \text{ km/s}} = \frac{15.6\pi \times 10^8}{13} = 1.2\pi \times 10^8 \text{ s}$$

To find the number of Earth orbits completed in Jupiter's orbital time:

$$\text{number of orbits} = \frac{T_J}{T_E} = \frac{1.2\pi \times 10^8}{\pi \times 10^7} = 1.2 \times 10 = 12$$

$$\text{number of orbits} = \boxed{12}$$